HOW MANY EQUATIONS DOES IT TAKE TO DEFINE AN ALGEBRAIC SET?

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The purpose of this note is to prove the following, as an exercise in commutative algebra.

Theorem 1 (Storch, Eisenbud-Evans). *Let $X \subset \mathbb{A}^n$ be an algebraic set. Then*

$$X = V(f_1, \dots, f_r)$$

for some $r \leq n$.

We first give an algebraic reformulation.

Proposition. *If $I \subset k[x_1, \dots, x_n] = S$ is an ideal, then there are $f_1, \dots, f_r \in R$, $r \leq n$ such that*

$$\sqrt{I} = \sqrt{(f_1, \dots, f_r)}$$

We say in this case that I is "generated up to radical" by $f_1, \dots, f_r$. We first prove the following statement, with weaker hypotheses and a weaker conclusion.

Proposition. *Let S be a Noetherian ring of dimension d. Then $I \subset S$ is generated up to radical by at most $d+1$ elements.*

Proof. Let $\mathfrak{N}$ be the nilradical of S. This is the intersection of all the primes of S, or equivalently, the intersection of all the minimal primes. If $I \subset \mathfrak{N}$ then $\sqrt{I} = \mathfrak{N}$, so I is generated up to radical by 0 elements. Otherwise, let

$$\mathfrak{N} = \mathfrak{p}_1 \cap \cdots \cap \mathfrak{p}_N$$

be a primary decomposition of $\mathfrak{N}$, so the $\mathfrak{p}_i$ are exactly the minimal primes. We are assuming that $I \not\subset \mathfrak{N}$, and we may assume that for $i \leq s$, $I \not\subset \mathfrak{p}_i$. By prime avoidance, we can choose $f \in I$ not contained in $\mathfrak{p}_1, \dots, \mathfrak{p}_s$. Now pass to $R = S/(f)$, $IR \subset R$. Consider the quantity $\delta(I)$, defined to be the maximum dimension of a minimal prime not containing I. This is a kind of coarse form of codimension; geometrically, it is the maximum dimension of an irreducible component of the ambient space that $X = V(I)$ meets in a proper subset. We have $\delta(IR) < \delta(I)$ by construction, so we can perform induction on this quantity. The base of the induction is to when $\delta(I) = -1$, in which case I is contained in the nilradical, the first case we considered. Geometrically, the algorithm can be described as follows. Examine the irreducible components of the ambient space that X meets properly, and find a function vanishing on X and not vanishing identically on any of these components. Then intersect everything with the hypersurface $V(f)$. This reduces the dimension of the components not contained in X until they are points, at which point the next hypersurface can cut out X exactly, avoiding all of them. This expresses X as an intersection of at most $d+1$ hypersurfaces. ☐

Now to get the more refined result for $k[x_1, \dots, x_n]$ quoted above, we prove the following.

Proposition. *Suppose $S = R[x]$ is of dimension d, where R is of dimension $e = d - 1$. Then any $I \subset S$ is generated up to radical by at most d elements.*

Proof. We prove this by induction on e. As a first step, we get rid of nilpotents in R and S. Let $\mathfrak{N}$ be the nilradical of S. Then $\sqrt{I} \supset \mathfrak{N}$, so we may as well assume that $\mathfrak{N} \subset I$. Then I corresponds to an ideal of $S/\mathfrak{N}$. Assuming that this ideal is generated up to radical by at most d elements, so is I. Thus, we can reduce the general case for a given value of e to the case that S is reduced, and so R is reduced as well. In case $e = 0$, R is a product of fields, and S is a product of polynomial rings in a single variable.

$$S = \prod_i k_i[x]$$

Date: January 2025.

Let $I_i = I \cap k_i[x]$. Choose f_i a generator of I_i. Then $(f_i)_i \in S$ generates I. This completes the proof for $e = 0$.

Now we consider $e > 0$. Let $R[U^{-1}]$ be the total quotient ring of R, obtained by inverting all nonzerodivisors in R. Consider $(0) = \mathfrak{p}_1 \cap \cdots \cap \mathfrak{p}_N$ an irredundant primary decomposition of (0) in R. The following follows from the theory of primary decomposition, but we can give a direct proof.

Lemma 1. *An element $r \in R$ is a zerodivisor if and only if it is contained in one of the $\mathfrak{p}_i$.*

Proof. Suppose r is contained in $\mathfrak{p}_1, \ldots, \mathfrak{p}_k$. By prime avoidance, we can find an r' contained in $\mathfrak{p}_j$ for $j > k$, but not in any of the $\mathfrak{p}_1, \ldots, \mathfrak{p}_k$. Then $r \cdot r'$ is in the intersection of all the minimal primes, so is 0, but r' is nonzero, as it is not contained in some prime.

Suppose now that r is not contained in any of the $\mathfrak{p}_i$. Then if $r \cdot r' = 0$, it must be that r' is in every $\mathfrak{p}_i$, so r' is already 0. Thus r is not a zerodivisor. $\square$

We also give a direct proof of the following statement, which follows as $R[U^{-1}]$ is zero dimensional.

Lemma 2. *$R[U^{-1}]$ is a product of fields. Specifically, there is an isomorphism*

$$R[U^{-1}] \to \prod_j K(R/\mathfrak{p}_j)$$

Proof. We have an injective map

$$R \to \prod_j K(R/\mathfrak{p}_j)$$

Each nonzerodivisor is sent to a tuple of nonzero elements in each field, which is a unit in the product. Therefore, the map lifts to an injective map

$$R[U^{-1}] \to \prod_j K(R/\mathfrak{p}_j)$$

It remains to show that this is surjective. This will follow from prime avoidance. We can find an r_j such that $r_j \notin \mathfrak{p}_j$ but $r_j \in \mathfrak{p}_i$ for every $i \neq j$. Then r_i maps to a unit in $K(R/\mathfrak{p})$, so taking a multiple of r_i, we may assume that it maps to 1. Then for any

$$\left(\frac{f_j}{g_j}\right) \in \prod_j K(R/\mathfrak{p}_j)$$

we may first choose lifts $\widetilde{g_j}$ of the $\mathfrak{q}_j$ to R. These are defined up to $\mathfrak{p}_j$, so by adding appropriate r_i for $i \neq j$, we may assume that each $\widetilde{g_j}$ is not contained in $\mathfrak{p}_i$ for all i, so is a nonzerodivisor. Now let $\widetilde{f_j}$ lift f_j. Then

$$\sum_j \frac{\widetilde{f_j}}{\widetilde{g_j}} \cdot r_j \longmapsto \left(\frac{f_j}{g_j}\right)$$

This completes the proof of surjectivity. $\square$

Now we can as in the zero dimensional case choose a generator f of $I \cdot R[U^{-1}][t]$, which we may assume comes from I by clearing denominators. Now let $g_1, \ldots, g_N$ be some generators of I. For each g_i, we have an equation

$$g_i = \frac{q_i}{r_i} f$$

where $q_i \in R[t]$ and $r_i \in R$ is a nonzerodivisor. Letting $r = \prod_i r_i$, we have $rg_i \in (f)$ for each i, so $rI \in (f)$. Because r is a product of nonzerodivisors, i.e. elements of the complements of the minimal primes $\mathfrak{p}_i$, the same is true of r.

Now consider $J = I \cdot (R/(r))[x]$. The dimension of $R/(r)$ is less than that of R, so we may assume, appealing to induction, that J is generated up to radical in $I \cdot (R/(r))[x]$ by the images of elements $f_2, \ldots, f_d \in S$. Let $h \in I$. We then have an equation

$$h^m = \sum_{2 \leq j \leq d} p_j \cdot f_j + q \cdot r$$

where $p_j, q \in S$. Multiplying everything by h, we have

$$h^{m+1} = \sum_{2 \le j \le d} p'_j \cdot f_j + q \cdot (hr)$$

Using the fact that $h \in I$, we have $hr \in (f)$, so $q \cdot hr = q' \cdot f$ for some q'. Thus

$$h^{m+1} \in \sum_{2 \le j \le d} p'_j \cdot f_j + q' \cdot f \in (f, f_2, \ldots, f_d)$$

We conclude that I is generated up to radical by $f, f_2, \ldots, f_d$. □

Geometrically, we are considering $X \subset Y \times \mathbb{A}^1$. We then find a hypersurface $Z \subset Y$ and a hypersurface $W \subset Y \times \mathbb{A}^1$ such that $W \supset X$ and $X \cup [Z \times \mathbb{A}^1] \supset W$. In other words,

$$X \cup [Z \times \mathbb{A}^1] = W \cup [Z \times \mathbb{A}^1]$$

Why can we do this? Look at the image πX_i of each component X_i of X in Y. If πX_i is contained in a hypersurface Z_i, then we can take $W_i = Z_i \times \mathbb{A}^1$. Otherwise, X_i projects dominantly onto Y. Each such X_i is itself a hypersurface, $W_i = X_i$. Now take $Z = \cup_i Z_i$, $W = \cup_i W_i$.

Appealing to induction on dimension, we can express $X \cap [Z \times \mathbb{A}^1]$ as an intersection of hypersurfaces $W_2 \cap \cdots \cap W_d$, $W_i \subset Z \times \mathbb{A}^1$. We can express W_j as $\widetilde{W_j} \cap [Z \times \mathbb{A}^1]$, where $\widetilde{W_j}$ is a hypersurface in $Y \times \mathbb{A}^1$. Then we have

$$X = W \cap \widetilde{W_2} \cap \cdots \cap \widetilde{W_d}$$

We check this equality by checking it after intersecting with $Z \times \mathbb{A}^1$, and with the complement of this.